BLANCHARD WATERSHED MODELING AND ANALYSIS USING SWAT

GEOL 6160 WATERSHED ANALYSIS & MODELING

BRENDA ONGERA

MODELING HYDROLOGICAL RESPONSES AND FLOOD RISK IN THE BLANCHARD RIVER WATERSHED

INTRODUCTION

The Blanchard River watershed, covering over 770 square miles in northwestern Ohio, is a tributary of the Auglaize River in the Maumee River basin. Dominated by row crop agriculture, the watershed was severely affected by the August 2007 flood, which caused significant damage in Findlay and Ottawa.

This project aims to model the watershed's hydrological responses using the SWAT model, with a focus on improving flood risk management and water quality.

DATA AND METHOD

The SWAT model uses key data including a Digital Elevation Model (DEM) for topography, land use data highlighting row crop agriculture, soil data for runoff and infiltration, and climate data (precipitation and temperature) for hydrological simulations. Observed river flow data from the USGS gaging station at Findlay was also used for model calibration and validation, focusing on water flow, sediment transport, and nutrient cycling, while considering the effects of land use, soil properties, and climate.

RESULTS AND DISCUSSION

This section presents SWAT model results for the Blanchard River watershed, focusing on hydrological processes, flood risk, and the impact of land use and climate.

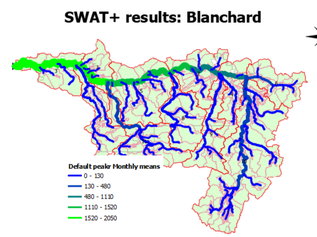

Figure 1: monthly Peak Flow

Figure 1 illustrates high water flow from the upstream areas (blue) towards the outlet (green), indicating a higher flood risk in the downstream region during heavy rainfall events.

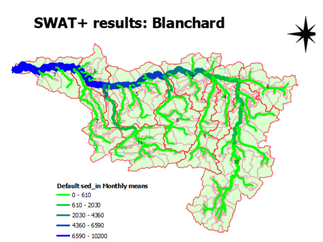

Figure 2: monthly sediment inflow

Figure 2 shows increased sediment inflow in the main channel (blue), which intensifies downstream towards the outlet, suggesting that sediment is transported from the upstream areas (green) and accumulates near the river's mouth.

PRECIPITATION AND DISCHARGE COMPARISON

In the this watershed, there is a correlation between rainfall and river flow, with heavy rainfall events causing significant spikes in discharge (see figure 3)

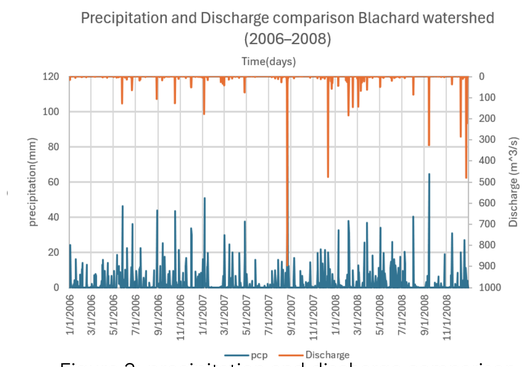

Figure 3: precipitation and discharge comparison

This suggests that the watershed has limited capacity to store water, leading to rapid runoff into the river. The most notable example is the August 2007 flood event, where intense rainfall led to dramatic increases in river discharge, particularly on August 21–22, resulting in discharge levels of up to 896 cfs and indicating that the watershed's capacity to absorb water was overwhelmed, causing flooding

IMPACT OF CN2 VARIATION ON RUNOFF SIMULATION

Varying Curve Number (CN2) by ±20% significantly impacts runoff (see figure 4). A 20% increase in CN2 results in higher and sharper runoff peaks due to reduced infiltration, while a 20% decrease leads to lower peaks and slower runoff, indicating increased water retention. These results emphasize the sensitivity of runoff generation to CN2 changes, highlighting the need for accurate calibration in hydrological models to predict flood behavior effectively

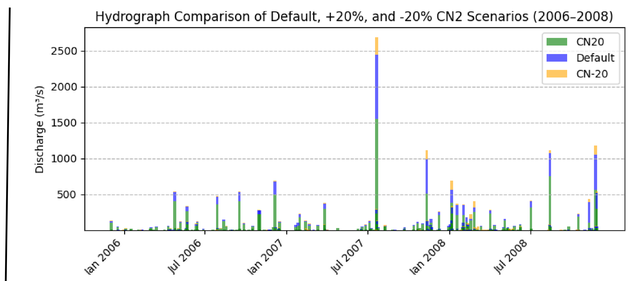

figure 4: a grapgh showing the impact of CN2 variation on runoff

OBSERVED VS. SIMULATED FLOWS AT FINDLAY USGS GAGING STATION

Figure 5 below shows that the model underestimated the flow, suggesting a need for calibration by adjusting the Curve Number (CN2) value. I would increase CN2 in small increments and re-run the simulation to improve flow predictions. This iterative process would involve comparing the observed and simulated hydrographs visually and using statistical measures like the Root Mean Square Error (RMSE) to quantify improvements.

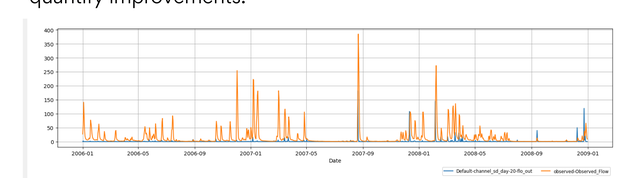

Figure 5: Observed Vs simulated flows

CONCLUSION

In conclusion, this project used the SWAT model to simulate hydrological responses to climatic variability, focusing on the August 2007 flood. The results highlighted the model's ability to capture flow dynamics and the sensitivity of parameters like Curve Number (CN2) in runoff generation. Adjusting CN2 provided valuable insights into calibration strategies, emphasizing the importance of hydrological modeling for flood management and watershed planning

BGSU School of Earth, Environment and Society
BOWLING GREEN STATE UNIVERSITY